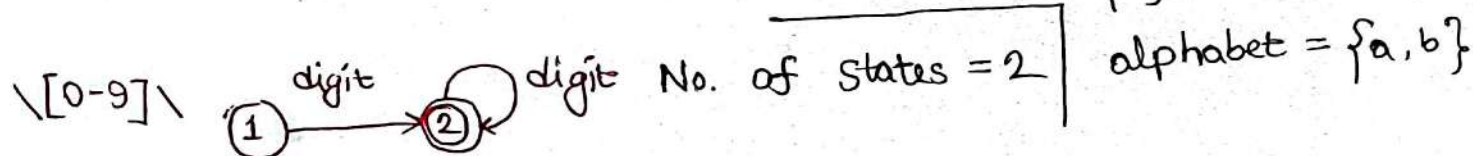
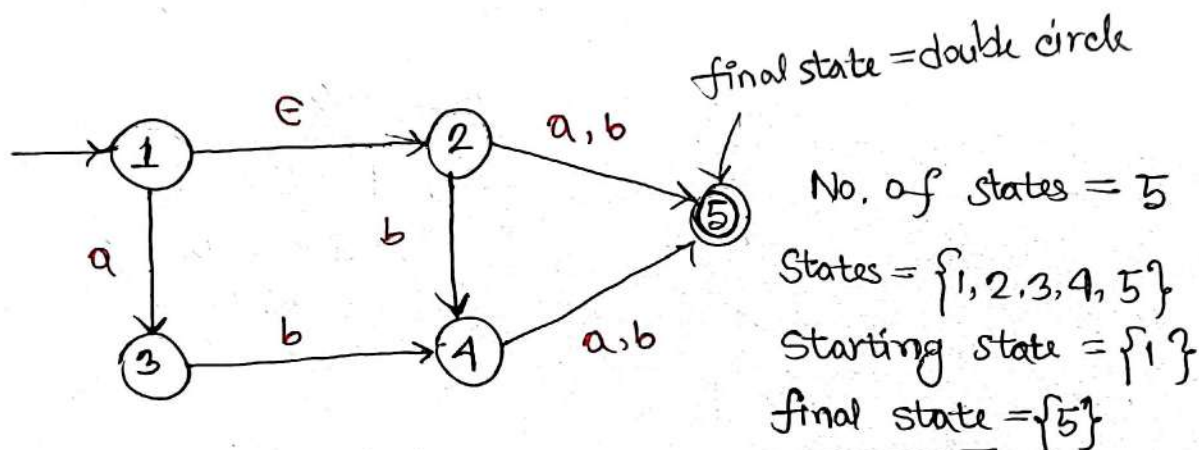




#NFA to DFA using Subset Construction

NFA



DFA state

Non-deterministic Finite Automata (NFA)

Deterministic Finite Automata (DFA)

STEP-1:

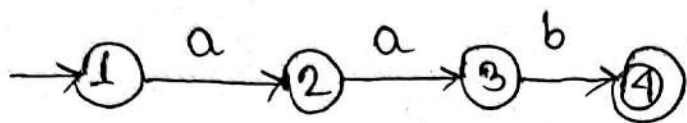
Transition table for NFA:

No. of state = No. of rows

state	a	b	ϵ
1	3	-	2
2	5	4, 5	-
3	-	4	-
4	5	5	-
5	-	-	-

1) Bayan / ②

aab



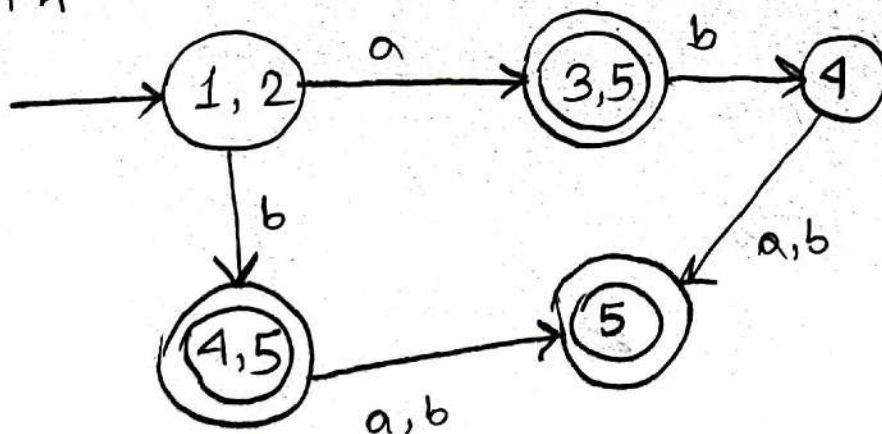
STEP-2:

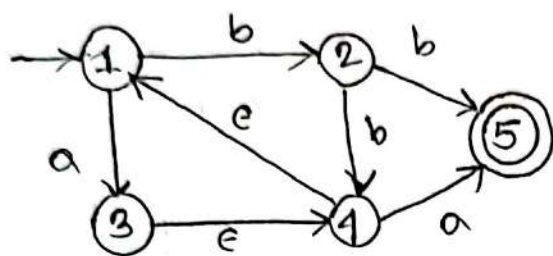
Transition Diagram for DFA

State	a	b
1, 2	3, 5	4, 5
3, 5	—	4
4, 5	5	5
4	5	5
5	—	—

empty string
 $1 \xrightarrow{\epsilon} 2$
 ∴ 1 delivered 'ε' to 2
 1 & 2 will be merged

DFA





STEP-1

Transition table for NFA:

State	a	b	ε
1	3	2	-
2	-	4,5	-
3	-	-	4
4	5	-	1
5	-	-	-

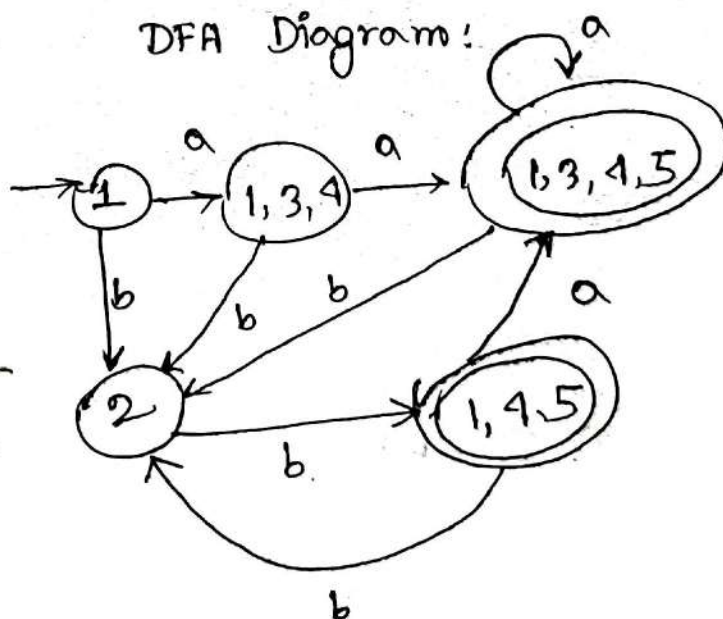
STEP-2

Transition diagram for DFA:

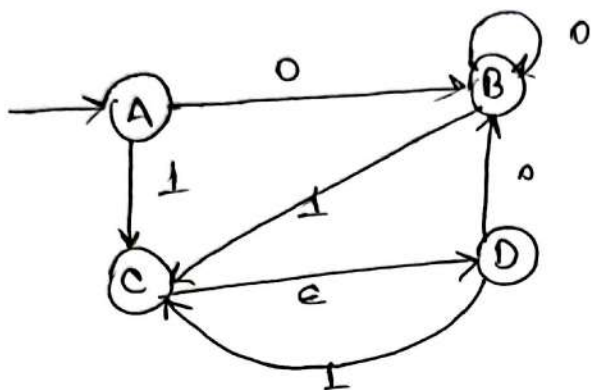
State	a	b
1	1,3,4	2
1,3,4	1,3,4,5	2
2	-	1,4,5
1,3,4,5	1,3,4,5	2
1,4,5	1,3,4,5	2

STEP-3

DFA Diagram:



Shahar / (4)



0 0
0 0 0
0 0 0
0 0 0

STEP 1: Transition table for NFA:

State	0	1	ϵ
A	B	C	
B	B	C	
C	—	—	D
D	B	C	

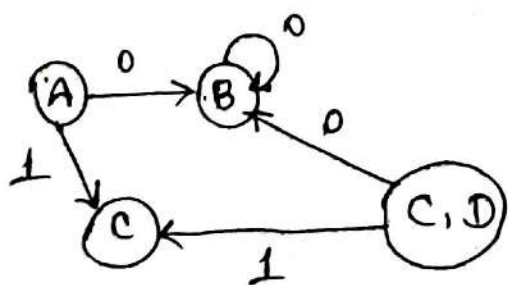
LL(1)

Grammar

STEP 2: Transition diagram for DFA:

State	0	1
A	B	C
B	B	C
C, D	B	C

Identify LL(1) grammar



$P \rightarrow 1 | c | nTS$

$Q \rightarrow P | aS | bQc$

$R \rightarrow b | \epsilon$

$S \rightarrow c | Rn | \epsilon$

$T \rightarrow Sq | R$